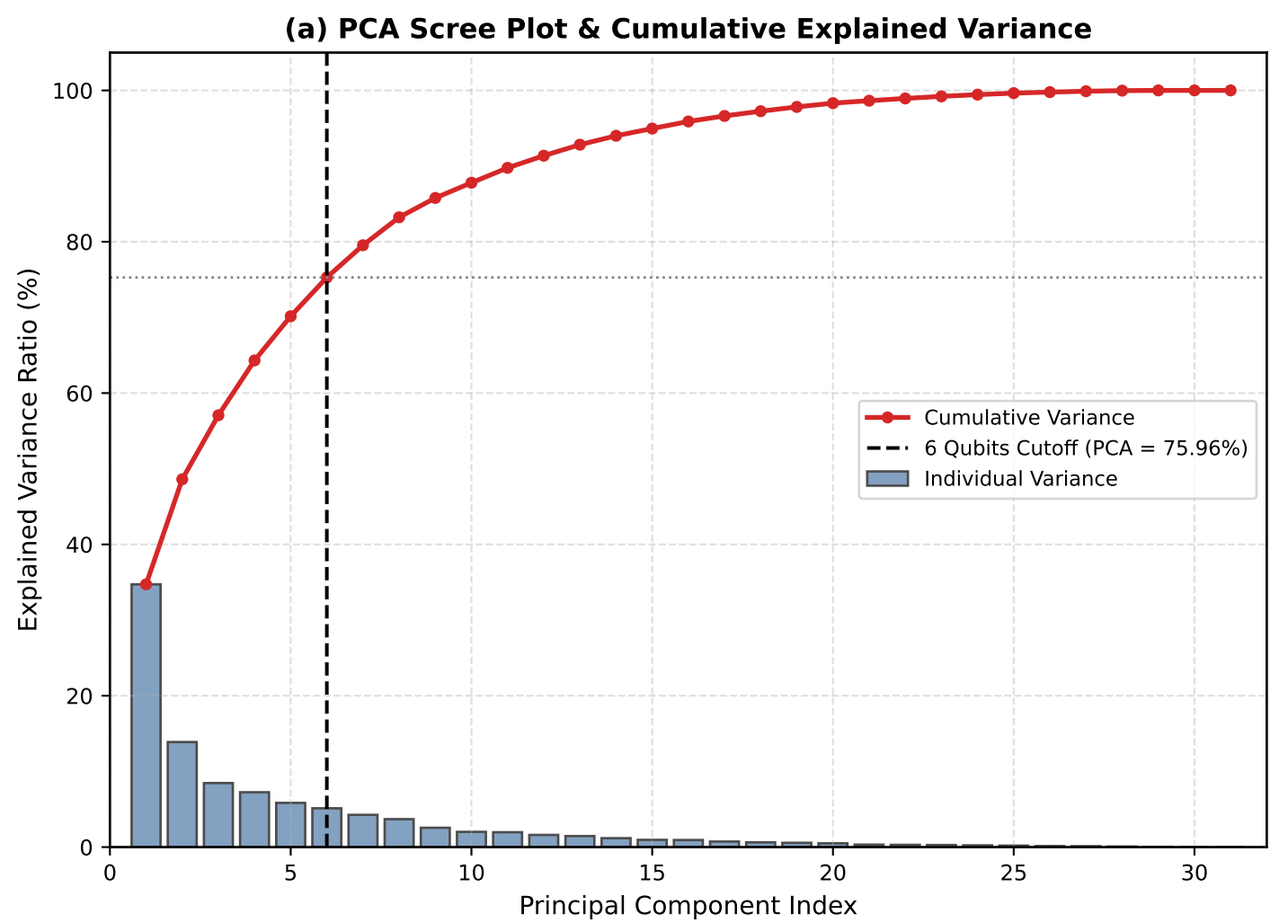
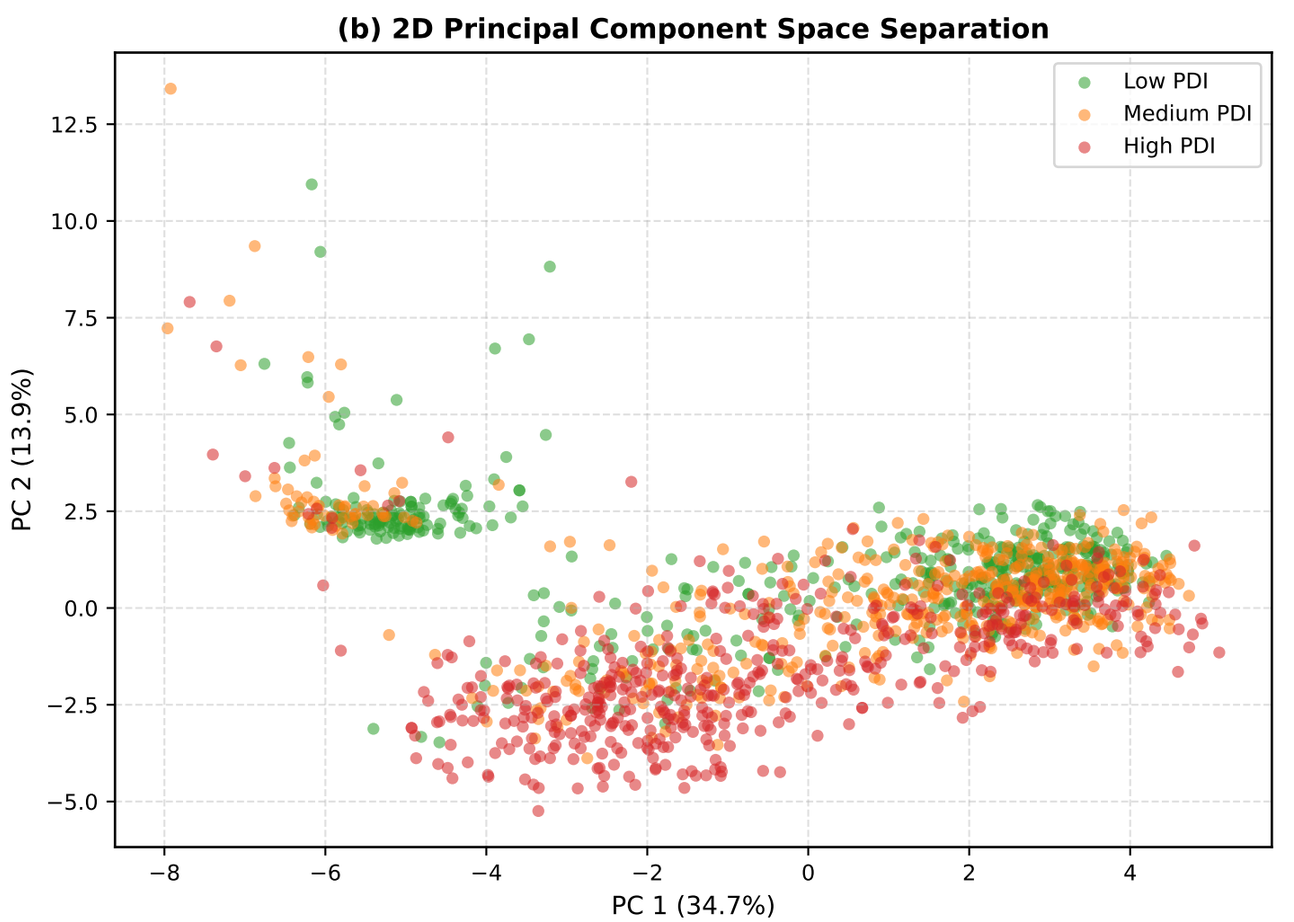
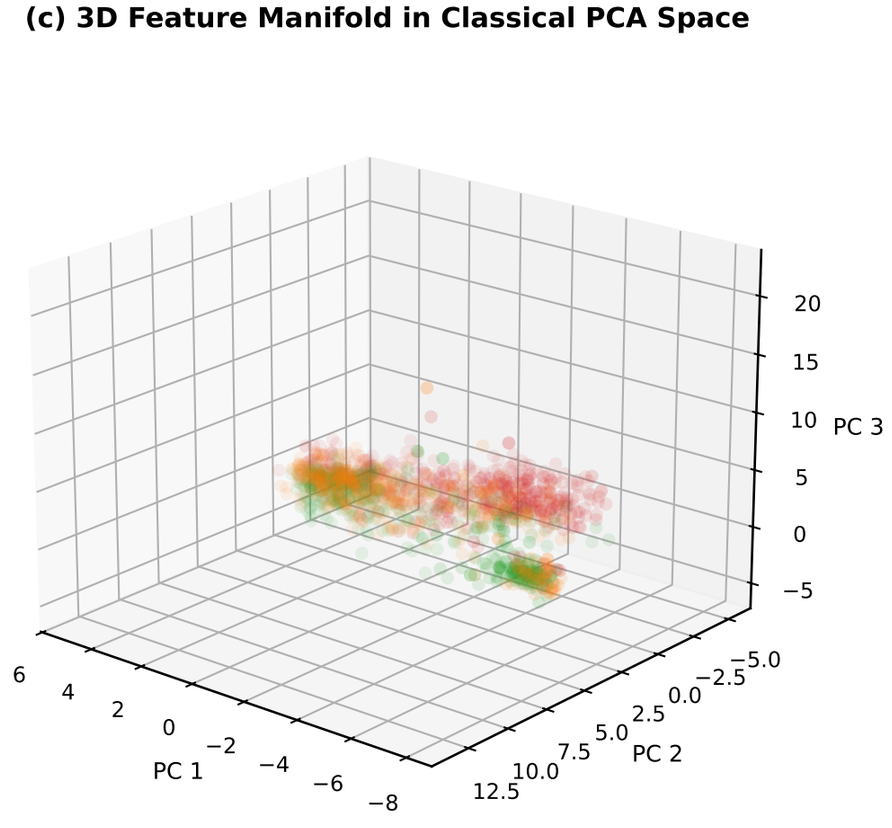
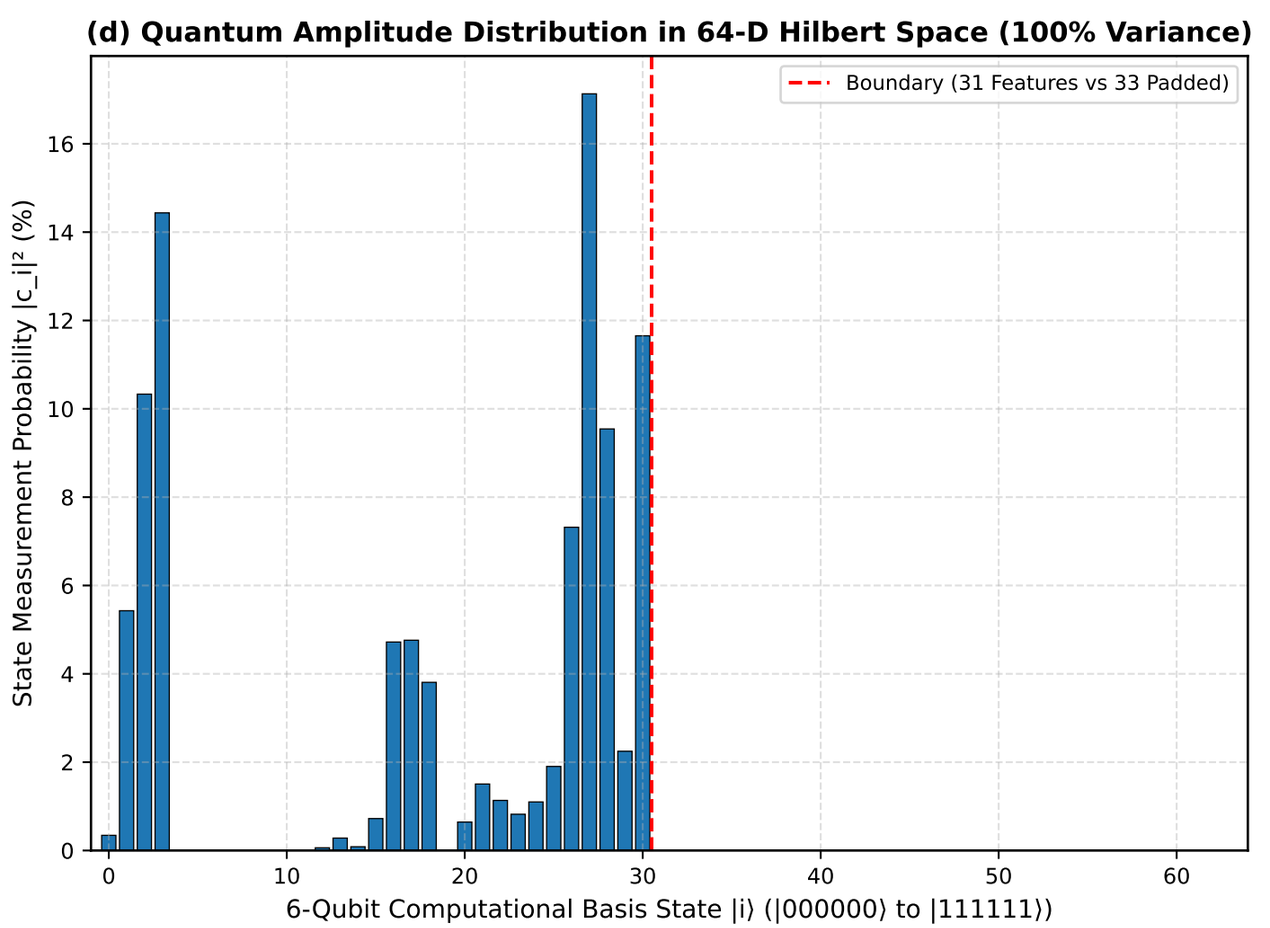

(a) PCA Scree Plot & Cumulative Explained Variance
Explained Variance Ratio (%)
Principal Component Index
Cumulative Variance
6 Qubits Cutoff (PCA = 75.96%)
Individual Variance


(b) 2D Principal Component Space Separation
PC 2 (13.9%)
PC 1 (34.7%)
Low PDI
Medium PDI
High PDI


(c) 3D Feature Manifold in Classical PCA Space
PC 1
PC 2
PC 3


(d) Quantum Amplitude Distribution in 64-D Hilbert Space (100% Variance)
State Measurement Probability |c_i|² (%)
6-Qubit Computational Basis State |i⟩ (|000000⟩ to |111111⟩)
Boundary (31 Features vs 33 Padded)